THEME 4: APPLICATIONS OF DIFFERENTIATION

Unit 4.1 Maximum and minimum values

Source: Stewart, Section 4.1 pp 276 - 287.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an absolute maximum of a function, an absolute minimum of a function, a local maximum of a function, a local minimum of a function and a critical number of a function;
2. be able to write down and use the Extreme Value Theorem, Fermat's Theorem;
3. know that the converse of Fermat's Theorem is false;
4. be able to find the critical numbers of a function;
5. be able to use the Closed Interval Method to find the absolute extremes of a continuous function on a closed interval;
6. be able to find the absolute and local extremes of a function if the graph of the function is given.

Remarks

1. You do not have to prove any theorems in this unit.
2. It is important to remember that if c is a critical number it does not mean that $f(c)$ is a local extreme. You have to use the tests given in Unit 4.3 to test whether it is a local extreme or not.
3. Omit Example 9.

Assigned Problems

Stewart, Exercises 4.1 p 286: Numbers 3, 5, 9, 19, 20, 21, 27, 31, 33, 41, 45, 55, 59, 61, 63, 64, 65.

Unit 4.2 The Mean Value Theorem

Source: Stewart, Section 4.2 pp 290 - 296.

Learning objectives

On completion of this unit you should

1. be able to write down Rolle's Theorem and be able to apply it in problems;
2. be able to write down the Mean Value Theorem and be able to apply it in problems.

Remarks

1. You do not have to prove any of the theorems in this unit.
2. Omit Example 6.

Assigned Problems

Stewart, Exercises 4.2 p 295: Numbers 3, 6, 9, 12, 17, 21, 31, 33.

Unit 4.3 How the derivative affects the shape of the graph

Source: Stewart, Section 4.3 pp 296 - 309.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a function that is increasing on an interval;
2. be able to write down and use the definition of a function that is decreasing on an interval;
3. be able to use the Increasing/Decreasing Test to find the interval(s) on which a function is increasing and the interval(s) on which a function is decreasing;
4. be able to use the First Derivative Test to find the local extremes of a function;
5. be able to write down and use the definition of a function that is concave upward on an interval;
6. be able to write down and use the definition of a function that is concave downward on an interval;
7. be able to use the Concavity Test to find the interval(s) on which a function is concave downward and the interval(s) on which a function is concave upward;
8. be able to write down and use the definition of an inflection point;
9. be able to use the second derivative to find the inflection point(s) of a function;
10. be able to use the Second Derivative Test to find the local extremes of a function.

Remarks

1. The definition of an increasing / a decreasing function is on p 17.
2. Leave Examples 6, 7 and 8 for now. We will do curve sketching in Unit 4.5.

Assigned Problems

Stewart, Exercises 4.3 p 305: Numbers 1, 5, 7, 9, 15, 16, 19, 25, 26, 29, 40, 53, 57, 84.

Unit 4.4 Indeterminate forms and L'Hospital's rule

Source: Stewart, Section 4.4 pp 309 - 319.

Learning objectives

On completion of this unit you should

1. be able to explain what indeterminate limit forms are;
2. be able to write down the seven different types of indeterminate forms;
3. be able to identify when a limit is of indeterminate form;
4. be able to use the rule of L'Hospital to find certain limits of indeterminate forms.

Remark

You always have to write down when a limit has an indeterminate form and you have to indicate that you use L'Hospital's rule, e.g.,

$$\begin{aligned} & \lim_{x\to\infty}\frac{e^x}{x^2} \qquad (\text{form } \frac{\infty}{\infty}) \quad \text{L'Hospital} \\ = & \lim_{x\to\infty}\frac{e^x}{2x} \qquad (\text{form } \frac{\infty}{\infty}) \quad \text{L'Hospital} \\ = & \lim_{x\to\infty}\frac{e^x}{2} \\ = & \infty. \end{aligned}$$

Assigned Problems

Stewart, Exercises 4.4 p 316: Numbers 7, 9, 11, 15, 17, 25, 29, 49, 51, 53, 56, 57, 58, 59, 60, 65, 77.

Unit 4.5 Curve sketching

Source: Stewart, Section 4.5 pp 320 - 329.

Learning objectives

On completion of this unit you should be able to sketch the graph of a given function.

Remark

To sketch the graph of a function, you have to follow the guidelines given in this section, i.e., you have to

1. find the domain of the function;

2. if possible, find the intercepts of the function with the $x-$axis and with the $y-$axis;
3. determine whether the function is even, odd or periodic, or none of these;
4. find the horizontal asymptotes (if any) of the function;
5. find the vertical asymptotes (if any) of the function;
6. find the interval(s) on which the function is increasing and the interval(s) on which the function is decreasing;
7. find the local extremes of the function (use number 6);
8. find the interval(s) on which the function is concave upward and the interval(s) on which the function is concave downward;
9. find the inflection points of the function;
10. use the information above to sketch the graph of the function.

Remark: Do examples 6, 7 and 8 on pp 302 - 305.

Assigned Problems

1. Stewart, Exercises 4.5 p 327: Numbers 7, 13, 27, 41, 52, 53.
2. Stewart. Exercises 4.3 p 305: Numbers 35, 37, 38.

Unit 4.6 Optimization problems

Source: Stewart, Section 4.7 pp 336 - 349.

Learning objectives

On completion of this unit you should

1. know what is meant by an optimization problem;
2. be able to convert a word problem into a mathematical optimization problem by setting up the function that has to be minimized or maximized;
3. be able to use the methods in this theme to find absolute extreme values;
4. be able to solve optimization problems such as described in the examples.

Remarks

1. The main task in this section is to convert a word problem into a mathematically formulated problem.
2. Making a sketch is invaluable.

Assigned Problems

Stewart, Exercises 4.7 p 342: Numbers 1, 3, 5, 9, 21.
